Lab Sheet 05 – Question 03: Online Bookshop

SE1020 – Object Oriented Programming

(a) Nouns that can be potential classes

online bookshop, products, books, magazines, CDs, information, price, title, author, number of pages, artists, type, registered customers, customer, orders, order, shipments, store, personal customers, business customers, payments, delivery, invoice, credit card, contact information, personal information, authentication, system, reports, report.

(b) Rejecting nouns using the five rules

Noun	Decision	Rule applied / Reason
online bookshop, store, system	Rejected	Irrelevant / Implementation construct – the system itself is not modelled as a class
information	Rejected	Vague – no clear meaning as a class
price, title	Rejected	Attributes of Product
author, number of pages	Rejected	Attributes of Book
artists	Rejected	Attribute of CD
type	Rejected	Attribute (e.g. type of CD / magazine)
contact information, personal information	Rejected	Attributes of Customer
authentication	Rejected	Irrelevant – an operation/process, not an object
registered customers	Rejected	Adjective class – "registered" only describes Customer
delivery	Rejected	Redundant – represented by Shipment
payments	Rejected	Redundant form – same as Payment (kept as Payment)
products, orders, shipments, reports	Rejected (plural)	Redundant – same as Product, Order, Shipment, Report

Final set of classes

1. **Product** (super class) – with subclasses **Book**, **Magazine**, **CD**
2. **Customer** (super class) – with subclasses **PersonalCustomer**, **BusinessCustomer**
3. **Order**
4. **Shipment**
5. **Payment**
6. **Invoice**
7. **CreditCard**
8. **Report**